

How to use this sheet: find the words the question uses (in bold), follow the numbered steps, copy the layout of the example. For every number answer write: formula → plug in → arithmetic → boxed answer with units. Relative frequencies as decimals (0.3000). Probabilities to 4 decimals.

SYMBOL DECODER (read this first)

- Σ ("sigma," big) = "add them all up." Σx = add all the values.
- x = one data value. N = how many values in the POPULATION. n = how many in the SAMPLE.
- μ ("mu") = population mean (average). $\bar{x}$ (" x -bar") = sample mean. Same math, different letter.
- σ ("sigma," small) = population standard deviation. σ^2 = population variance (= σ squared).
- s = sample standard deviation. s^2 = sample variance.
- p = population proportion; $\hat{p}$ ("p-hat") = sample proportion (a percent written as a decimal).
- M = median (middle value). $Q_1 / Q_2 / Q_3$ = quartiles (25% / 50% / 75% marks; Q_2 = median).
- $IQR = Q_3 - Q_1$. LF / UF = lower / upper fence (outlier cutoffs). z = z -score (how many SDs from the mean).
- RF = relative frequency = part ÷ total. $^\circ$ = degrees (pie chart). $\sqrt{}$ = square root. $()^2$ = square it.
- Greek letters + capital N = POPULATION (everyone). Regular letters + small n = SAMPLE (some of them).

CHAPTER 1 — COLLECTING DATA

If asked: "identify the population / sample / individuals / parameter / statistic" ★Q1

- **Population** = the WHOLE group you want to know about. Look at what the study is about: "all people who could follow the diet," "all US adults with high blood pressure."
- **Sample** = the smaller group actually measured. Look for the number: "80 participants," "165 adults."
- **Individual** = ONE member (one participant, one student).
- **Parameter** = a number describing the whole population. Usually UNKNOWN → describe it in words: "the true average cholesterol drop for everyone on the diet (μ)."
- **Statistic** = a number computed from the sample = the number they GIVE you: "average drop of 15 points ($\bar{x} = 15$)."
- Memory: Parameter ↔ Population; Statistic ↔ Sample. If a number is stated for ALL (e.g., 0.849 of all students), that number IS the parameter.
- Statistics process: (1) state the question, (2) collect data, (3) describe it (graphs, averages), (4) inference (use the sample to say something about the population).

If asked: "qualitative or quantitative / discrete or continuous / level of measurement" ★Q2 — answer ALL THREE

Step 1 — Qualitative or Quantitative? - Qualitative (categorical) = words, labels, categories: car type, phone brand, yes/no, "strongly agree," eye color. **ALSO rankings (1st, 2nd, 3rd)** and number-labels (ZIP code, ID) — they look like numbers but are labels. - **Quantitative** = actual numbers you can average: temperature, number of pets, income, hours of sleep, height.

Step 2 — Discrete or Continuous? (only for quantitative; if qualitative write "Not applicable (not quantitative)") - **Discrete** = you COUNT it; whole numbers with gaps ("number of...", "how many..."). Can't have 2.5 pets. - **Continuous** = you MEASURE it; any value in a range, decimals allowed: temperature, time/hours, weight, height, distance, income (\$), age, grade (%).

Step 3 — Level of measurement: ask these questions IN ORDER 1. Can the values be put in a meaningful order? **NO** → **Nominal** (car type, phone brand, yes/no, ZIP, gender). 2. Yes, order. Are the gaps between values equal/meaningful? **NO** → **Ordinal** (rankings, small/medium/large, strongly agree... disagree, letter grade, education level,

young/middle/old, star rating). 3. Yes, gaps meaningful. Does 0 mean "none at all"? **NO** → **Interval** (temperature $^\circ F/^\circ C$ — 0° is not "no temperature," 80° is not twice 40° ; calendar year, clock time, SAT score). 4. Yes, 0 = none. → **Ratio** (any count, money, time, weight, height, age, distance, grams, % grade).

Shortcuts: qualitative → always Nominal or Ordinal. Quantitative → always Interval or Ratio, and it is **Ratio unless it's temperature/year/clock time (Interval)**.

Worked (★Q2): (a) Transmission type → Qualitative; Not applicable; Nominal. (b) Number of pets → Quantitative; Discrete; Ratio. (c) Temperature in $^\circ F$ → Quantitative; Continuous; Interval. (d) Ranking of top 5 players → Qualitative; Not applicable; Ordinal. More: education level → Qual, N/A, Ordinal. Grams of whole grains per day → Quant, Continuous, Ratio. Number of vending machines / students → Quant, Discrete, Ratio. Prepared for class (yes/no) → Qual, N/A, Nominal. Age in years / income in \$ / hours of sleep / algebra grade % → Quant, Continuous, Ratio. Internet provider / phone type → Qual, N/A, Nominal. "Working with numbers upsets me" (agree...disagree) → Qual, N/A, Ordinal.

If asked: "observational study or experiment? explanatory / response variable? can we claim causation?" ★Q3

- **Explanatory variable** = the thing that might cause a change (the input/cause): screen time, the drug, type of music lesson.
- **Response variable** = the outcome measured (the result): GPA, blood pressure, IQ score.
- **Observational study** = researchers only WATCH, ASK, SURVEY, RECORD, TRACK, or compare groups that already exist. Nobody is told what to do.
- **Designed experiment** = the researcher ASSIGNS the treatment: "randomly assigned to groups," "given the drug / a placebo," "control group."
- **CAUSATION RULE: observational → only association, answer "No" to causation. Experiment → can claim causation.** Why: a **lurking variable** (something not measured, like sleep or study habits) could be the real cause.
- Worked (★Q3, survey of screen time and GPA): (a) Observational (only surveyed). (b) Explanatory = daily screen time. (c) Response = GPA. (d) Causation? **No**. Notes: kids randomly assigned to xylophone/singing/listening → experiment; children given milk with/without xylitol → experiment; surveys of new mothers / faculty / teens tracked for years → observational.
- **Census** = measure the ENTIRE population. **Simple random sample (SRS)** = every possible group of n has the same chance. To take one: (1) list everyone (the "frame") and number them 1 to N ; (2) generate n random numbers; those people are the sample (435 House members → number 1–435, generate 5 random numbers).

If asked: "what type of bias?" ★Q4 — ask three questions in order

1. **Who got asked?** If the wrong or only part of the group was asked → **SAMPLING bias**. ("Survey sent only to headquarters," "polled only its own readers," "people at the mall," "online poll people volunteer for.")
2. **Did the chosen people answer?** If many didn't and they differ from those who did → **NONRESPONSE bias**. ("Only 20% responded, mostly the unhappy ones," "refused," "hung up.")
3. **Are the answers true?** If people lie, give "safe" answers, or the question pushes an answer → **RESPONSE bias**. ("Answers management wants to hear," leading/loading question, embarrassing topic, mistyped data.) - **Nonsampling error** = mistakes from those biases (possible even in a census). **Sampling error** = the natural difference between a sample's number and the population's number just because you didn't measure everyone. Literary Digest 1936: sampling bias (only readers, car and phone owners) + nonresponse (only 2.38 million of 10 million replied) → predicted the wrong winner.

CHAPTER 2 — TABLES AND GRAPHS

If given a table of categories with counts: "fill in relative frequency / degrees / percent" ★Q5

- **Frequency** = the count. **Relative frequency** = $\text{count} \div \text{total}$ (decimal; all RFs add to 1). **Degrees** = $RF \times 360$ (all add to 360). **Percent** = $RF \times 100$.
- Backwards: **count** = $RF \times \text{total}$; missing $RF = 1 - (\text{the others})$; missing count = $\text{total} - (\text{the others})$.
- Worked (★Q5, total $60+50+40+30+20 = 200$): Sports $60/200 = 0.3000$, $0.3000 \times 360 = 108^\circ$. Movies $50/200 = 0.2500$, 90° . Shopping $0.2000 \times 360 = 72^\circ$. Reading $30/200 = 0.1500$, 54° . Other 0.1000 , 36° . Check: $108+90+72+54+36 = 360 \checkmark$. (b) Most popular = biggest count → Sports. (c) Reading = $0.15 \rightarrow 15\%$. (d) Sports or Movies = $(60+50)/200 = 110/200 = 0.55 = 55\% > 50\% \rightarrow \text{True}$.
- **Bar graph:** one bar per category, height = count; gaps between bars; **axis must start at 0, equal widths, 2-D**. **Pareto** = bars tallest → shortest. **Pie chart:** slice angle = $RF \times 360^\circ$; use for parts of a whole; use bars to compare specific categories. Comparing two groups of different sizes → compare relative frequencies, never raw counts.

If given classes like 40–49, 50–59 ...: "class limits / class width / how many / what percent" ★Q6

- **Lower class limit** = first number of the class (60 in 60–69). **Upper class limit** = last number (69).
- **Class width** = **start of next class – start of this class**: $50 - 40 = 10$. (NOT $49 - 40 = 9$ — classic trap.)
- **How many in a range** → add the counts of those classes. **What percent** → (those counts ÷ total) $\times 100$. **Most frequent class** → the biggest count; answer with the interval (70–79).
- **Histogram** = bar graph of these classes, bars touching. **Open-ended** class = " < 75 " or "300 or more."
- Worked (★Q6, 40 students; 40–49: 4, 50–59: 6, 60–69: 10, 70–79: 12, 80–89: 8): (a) 3rd class 60–69 → lower 60, upper 69. (b) width $50 - 40 = 10$. (c) 70 or higher: $12 + 8 = 20 \rightarrow 20/40 = 50\%$. (d) below 60: $4 + 6 = 10$. (e) biggest count 12 → **70–79**.

If asked: "what shape?" ★Q7 — look where the long thin TAIL points

- **Bell-shaped:** hump in the middle, mirror image. Heights, weights, IQ, most test scores.
- **Uniform:** flat, all values equally common. Fair die rolls, random digits.
- **Skewed RIGHT:** tail to the right = a few unusually BIG values. Salaries (a few executives earn millions), house prices, waiting times (most short, a few very long).
- **Skewed LEFT:** tail to the left = a few unusually SMALL values. Easy exam where most score high and a few bomb.
- Worked (★Q7): salaries → skewed right; heights of 10-year-olds → bell-shaped; die rolled 200 times → uniform; waiting times → skewed right.
- Link to Ch 3: skewed right → mean $>$ median; skewed left → mean $<$ median; symmetric → mean $\approx$ median.

Other graphs

- **Stem-and-leaf:** stem = all digits but the last, leaf = last digit ($56 \rightarrow 5 | 6$). List every stem in order (even empty ones), leaves in order, add a key. Scores 56, 60, 66, 68, 80, 82, 82, 84, 88, 90, 92, 94, 98, 98 → $5|6 \ 6|0 \ 6 \ 8 \ 7| \ 8|0 \ 2 \ 2 \ 4 \ 8 \ 9|0 \ 2 \ 4 \ 8 \ 8$.
- **Time-series plot:** time on the bottom, value on the side, dots connected. "Highest year" = highest dot.
- **Misleading graph:** vertical axis doesn't start at 0 (small differences look huge), bars of different widths, 3-D. Answer = name the flaw AND the fix ("start the axis at 0").

CHAPTER 3 — DESCRIBING DATA WITH NUMBERS

Running example = ★Q8–Q11: commute times of ALL 10 employees (so it's a **population**), already sorted: **18, 22, 25, 28, 30, 31, 33, 35, 36, 40** ($N = 10$).

If asked: "find the mean" ★Q8a, Q9a

Add all values, divide by how many. Population: $\mu = \Sigma x \div N$. Sample: $\bar{x} = \Sigma x \div n$. (Same math; use μ if the data is everyone / says "population," $\bar{x}$ if it says "sample.") Worked: $18+22+25+28+30+31+33+35+36+40 = 298 \rightarrow \mu = 298/10 = 29.8$ min. Sample of first 5 (18, 22, 25, 28, 30): $123/5 \rightarrow \bar{x} = 24.6$ min.

If asked: "find the median" ★Q8c

(1) SORT the data. (2) Count n . (3) n odd $\rightarrow$ the middle value (position $(n+1)/2$). n even $\rightarrow$ average the two middle values (positions $n/2$ and $n/2 + 1$). Worked: $n = 10$ (even) $\rightarrow$ 5th and 6th values are 30 and 31 $\rightarrow M = (30+31)/2 = 30.5$ min. Odd example: 76, 80, 84, **86**, 90, 92, 94 $\rightarrow M = 86$.

If asked: "find the mode" ★Q8d

The value that appears MOST. Two tie $\rightarrow$ both are modes ("bimodal": 7, 12, 12, 20, 20, 25 $\rightarrow$ 12 and 20). Nothing repeats $\rightarrow$ write "no mode". Worked: every commute time appears once $\rightarrow$ **no mode**.

If asked: "based on mean vs median, describe the shape" ★Q8e

The mean gets pulled toward the tail; the median doesn't. **mean > median $\rightarrow$ skewed right. mean < median $\rightarrow$ skewed left. mean $\approx$ median $\rightarrow$ symmetric.** Worked: mean 29.8 < median 30.5 $\rightarrow$ **skewed left (slightly)**. Write the comparison and the rule. **Resistant** = not thrown off by extreme values. Median and IQR are resistant; mean, range, SD are not. (Add a 70-min commuter to 5, 18, 23, 23, 26, 36, 43: mean jumps 24.9 $\rightarrow$ 30.5, median only 23 $\rightarrow$ 24.5.) Symmetric data $\rightarrow$ report mean & SD; skewed data $\rightarrow$ report median & IQR.

If asked: "find the range"

Biggest – smallest: $40 - 18 = 22$ min. (Easy, but one extreme value wrecks it.)

If asked: "find the variance and standard deviation" ★Q8b (population), Q9b (sample)

What it means: standard deviation (SD) $\approx$ how far the values typically are from the mean. Small SD = bunched up; big SD = spread out. Variance = SD squared (a middle step). **Why the recipe:** (value – mean) adds up to 0, so we SQUARE each difference (all positive), average the squares (= variance), then $\sqrt{}$ to undo the squaring (= SD). **THE RULE: "population" / all the data $\rightarrow$ divide by N . "sample" $\rightarrow$ divide by $n - 1$.** The word *sample* is your trigger. **Steps (draw this table!):** (1) find the mean. (2) column: value – mean. (3) column: square it. (4) add the squares. (5) divide by N (population) or $n - 1$ (sample) $\rightarrow$ **variance**. (6) $\sqrt{}$ $\rightarrow$ **standard deviation**. Check: the (value – mean) column should add to 0.

Worked ★Q8b (population, $\mu = 29.8$):

x	x – μ	(x – μ) ²
18	–11.8	139.24
22	–7.8	60.84
25	–4.8	23.04
28	–1.8	3.24
30	0.2	0.04
31	1.2	1.44
33	3.2	10.24
35	5.2	27.04
36	6.2	38.44
40	10.2	104.04
add	0 ✓	407.60

$\sigma^2 = 407.60 \div 10 = 40.76$ min². $\sigma = \sqrt{40.76} = 6.38$ min (6.3844 unrounded). (If the same 10 were called a sample: $407.60 \div 9 = 45.29$; $s = 6.73$.)

Worked ★Q9b (sample 18, 22, 25, 28, 30; $\bar{x} = 24.6$): differences –6.6, –2.6, 0.4, 3.4, 5.4 $\rightarrow$ squares 43.56, 6.76, 0.16, 11.56, 29.16 $\rightarrow$ add = **91.20** $\rightarrow s^2 = 91.20 \div 4 = 22.8$ min². $s = \sqrt{22.8} = 4.77$ min. **Calculator:** enter the data in 1-Var Stats. It shows TWO SDs: $\sigma x = \text{population } (\div N)$ and $Sx = \text{sample } (\div (n-1))$ — pick the right one. Variance = square the UNROUNDED SD ($6.3844^2 = 40.76$; never $6.4^2 = 40.96$). Still write the table — steps are required. TI-30XS/36X: [data] $\rightarrow$ enter in L1 $\rightarrow$ [2nd][data] $\rightarrow$ 1-Var Stats $\rightarrow$ CALC. Casio fx-991EX: MENU $\rightarrow$ Statistics $\rightarrow$ 1-Variable $\rightarrow$ enter $\rightarrow$ [OPTN] $\rightarrow$ 1-Variable Calc. Casio fx-300ES: MODE $\rightarrow$ STAT $\rightarrow$ 1-VAR $\rightarrow$ enter $\rightarrow$ AC $\rightarrow$ SHIFT 1 $\rightarrow$ Var. TI-84: STAT $\rightarrow$ Edit $\rightarrow$ L1; STAT $\rightarrow$ CALC $\rightarrow$ 1-Var Stats.

If asked: "Empirical Rule / 68–95–99.7 / between what two values / what percent between" ★Q11 (bell-shaped data only)

- About **68%** of values are within 1 SD of the mean ($\mu - \sigma$ to $\mu + \sigma$); **95%** within 2 SDs; **99.7%** within 3 SDs.
- Pieces: mean to 1 SD = **34%** (each side); 1 SD to 2 SD = **13.5%**; 2 SD to 3 SD = **2.35%**; beyond 3 SD = **0.15%**; above the mean = 50%.
- Steps:** (1) write μ and σ (round σ to 1 decimal if the question's numbers only match a rounded σ , and say so). (2) Compute the six landmarks: $\mu \pm 1\sigma$, $\mu \pm 2\sigma$, $\mu \pm 3\sigma$. (3) Find the question's numbers among the landmarks. (4) Add up the pieces between them.

Worked ★Q11 ($\mu = 29.8$; his numbers 23.4 and 36.2 = 29.8 ∓ 6.4 , so use $\sigma \approx 6.4$): $1\sigma \rightarrow 23.4$ and 36.2 ; 2σ (12.8) $\rightarrow 17.0$ and 42.6 ; 3σ (19.2) $\rightarrow 10.6$ and 49.0 .

10.6	17.0	23.4	29.8	36.2	42.6	49.0
2.35%	13.5%	34%	34%	13.5%	2.35%	

(a) 95% $\rightarrow$ 2 SDs: $29.8 - 12.8 = 17.0$ and $29.8 + 12.8 = 42.6 \rightarrow$ **between 17.0 and 42.6 min.** (b) 23.4 to 36.2 = 1 SD each side $\rightarrow$ **68%**. (c) 36.2 to 42.6 = from 1 SD to 2 SD above $\rightarrow (95 - 68) \div 2 = 13.5\%$. Quick answers: above $\mu + 1\sigma = 16\%$; above $\mu + 2\sigma = 2.5\%$; $\mu + 1\sigma$ to $\mu + 3\sigma = 13.5 + 2.35 = 15.85\%$; $\mu - 1\sigma$ to $\mu + 2\sigma = 34 + 34 + 13.5 = 81.5\%$. Notes Ex: waist $\mu = 48.5$, $\sigma = 4.8 \rightarrow 95\%$ between 38.9 and 58.1; 34.1 to $62.9 = \pm 3\sigma \rightarrow 99.7\%$; 53.3 to $62.9 = +1\sigma$ to $+3\sigma \rightarrow 15.85\%$. (Chebyshev: not tested.)

If asked: "z-score" or "who did relatively better?"

z = (value – mean) $\div$ SD = how many SDs from the mean; positive = above, negative = below, bigger z = relatively higher. ACT 27 (mean 18, SD 6): $z = (27 - 18)/6 = 1.50$. SAT 1200 (mean 1000, SD 200): $z = 200/200 = 1.00 \rightarrow$ the ACT student did relatively better. Commute 40 min: $z = (40 - 29.8)/6.38 = 1.60$. **Percentile:** "95th percentile" = 95% of the data is at or below you. Median = 50th percentile.

If asked: "quartiles / five-number summary" ★Q10a

(1) SORT. (2) **Q₂ = median** of everything. (3) **Q₁ = median of the LOWER half**; **Q₃ = median of the UPPER half**. If n is odd, leave the overall median OUT of both halves. (4) **Five-number summary = Min, Q₁, M, Q₃, Max**. Worked ($n = 10 \rightarrow$ halves of 5): lower 18, 22, **25**, 28, 30 $\rightarrow Q_1 = 25$; upper 31, 33, **35**, 36, 40 $\rightarrow Q_3 = 35$; $M = 30.5$. **Summary: 18, 25, 30.5, 35, 40**. Odd example: 76, 80, 84 | 86 | 90, 92, 94 $\rightarrow Q_1 = 80$, $Q_2 = 86$, $Q_3 = 92$. Even example with a low value: 32, 76, 80, 84 | 86, 90, 92, 94 $\rightarrow Q_1 = (76+80)/2 = 78$, $Q_2 = 85$, $Q_3 = (90+92)/2 = 91$.

If asked: "IQR / fences / outliers" ★Q10b,c

(1) **IQR = Q₃ – Q₁**. (2) **1.5 $\times$ IQR**. (3) **Lower fence = Q₁ – 1.5 $\times$ IQR**; **Upper fence = Q₃ + 1.5 $\times$ IQR**. (4) Any value BELOW the lower fence or ABOVE the upper fence is an **outlier**; compare the smallest and largest values to the fences. Worked: IQR = $35 - 25 = 10$; $1.5 \times 10 = 15$; **LF = 25 – 15 = 10**; **UF = 35 + 15 = 50**; $18 > 10$ and $40 < 50 \rightarrow$ **no outliers**. Example WITH an outlier: 32, 76, 80, 84, 86, 90, 92, 94 $\rightarrow Q_1 = 78$, $Q_3 = 91$, IQR = 13, $1.5 \times 13 = 19.5$, LF = 58.5, UF = 110.5 $\rightarrow 32 < 58.5 \rightarrow 32$ is an outlier.

If asked: "draw / read a boxplot"

(1) Five-number summary + fences. (2) Number line; box from Q_1 to Q_3 with a line at the median. (3) Whiskers from the box out to the smallest and largest values that are NOT outliers (not to the fences). (4) Each outlier = a separate *. Commute data (to scale, 2

chars = 1 min; no outliers, so whiskers reach 18 and 40):

18	25	30.5	35	40
----- ===== ===== -----				
Min	Q1	M	Q3	Max

With an outlier (32, 76 ... 94): box 78–91, line at 85, left whisker stops at 76 (the smallest NON-outlier), and 32 is drawn as *. **Reading shape:** median in the middle of the box + equal whiskers $\rightarrow$ symmetric; median toward the left + long right whisker or right-side * $\rightarrow$ skewed right (mirror $\rightarrow$ skewed left). Comparing two boxplots: compare medians (center), box widths/whiskers (spread), and outliers.

THE 10 WAYS PEOPLE LOSE POINTS ON THIS EXAM

1. Not sorting before the median/quartiles. 2. Dividing by n instead of $n - 1$ for a SAMPLE variance (Q9: $\div 4$, not $\div 5$). 3. Stopping at the variance when SD was asked — take the $\sqrt{}$ and write the units. 4. Calculator: Sx vs σx mixed up; squaring a rounded SD. 5. Odd n : putting the median into the halves. 6. Class width $49 - 40 = 9$ (it's $50 - 40 = 10$). 7. Relative frequency as 30% instead of 0.3000; degrees without $^\circ$. 8. Saying an observational study proves causation. 9. Naming skew by the hump instead of the TAIL. 10. Leaving "no mode" / "not applicable" blank — they ARE the answers. And: every number answer needs the steps written out.